RACE
Regular Analysis Through Continuous Exercise

Relations : RACE-01

Single Correct Question

1. Let $X = \{1,2,3,4,5\}$ and $Y = \{1,3,5,7,9\}$. Which of the following is not a relation from X to Y

(A) $R_1 = \{(x,y) | y = 2 + x, x \in X, y \in Y\}$

(B) $R_2 = \{(1,1), (2,1), (3,3), (4,3), (5,5)\}$

(C) $R_3 = \{(1,1), (1,3)(3,5), (3,7), (5,7)\}$

(D) $R_4 = \{(1,3), (2,5), (2,4), (7,9)\}$

MRL108

2. If $A = \{2,4,5\}$, $B = \{7, 8, 9\}$, then $n(A \times B)$ is equal to

(A) 6 (B) 9 (C) 3 (D) 0

MRL109

3. If A = {1, 2, 3}, B = {1, 4, 6, 9} and R is a relation from A to B defined by 'x is greater than y' $(x \in A, y \in B)$. Then range of R is-

(A) {1, 4, 6, 9} (B) {4, 6, 9} (C) {1} (D) none of these

MRL110

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4. The relation R is defined on the set of natural numbers as $\{(a,b): a = 2b\}$. Then R^{-1} is given by

(A) {(2, 1), (4, 2), (6, 3).....} (B) {(1, 2), (2, 4), (3, 6)....}

(C) R^{-1} is not defined (D) None of these

MRL111

5. The relation $R = \{(1,1), (2,2), (3,3), (1,2), (2,3), (1,3)\}$ on set $A = \{1,2,3\}$ is

(A) Reflexive but not symmetric (B) Reflexive but not transitive

(C) Symmetric and Transitive (D) Neither symmetric nor transitive

MRL112

6. Let $P = \{(x,y) | x^2 + y^2 = 1, x, y \in R\}$. Then P is

(A) Reflexive (B) Symmetric (C) Transitive (D) Anti-symmetric

MRL113

7. Let L be the set of all straight lines in the Euclidean plane. Two lines ℓ_1 and ℓ_2 are said to be related by the relation R if ℓ_1 is parallel to ℓ_2. Then the relation R is-

(A) Only Reflexive (B) Only Symmetric (C) Only Transitive (D) Equivalence

MRL114

8. Let X be a family of sets and R be a relation on X defined by 'A is disjoint from B'. Then R is-

(A) reflexive (B) symmetric (C) anti-symmetric (D) transitive

MRL115

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9. Let N denote the set of all natural numbers and R be the relation on $N \times N$ defined by (a, b) R (c, d) if ad $(b + c) = bc(a + d)$, then R is-

(A) Symmetric only (B) Reflexive only

(C) Transitive only (D) An equivalence relation

MRL116

10. **Statement-1 :** Let $a, b \in I$

$R_1 \rightarrow aR_1b \Leftrightarrow a + b$ is an even integer then R_1 is an equivalence relation.

$R_2 \rightarrow aR_2b \Leftrightarrow a - b$ is an even integer, then R_2 is an equivalence relation.

Statement-2 : A relation which is reflexive, symmetric and transitive is called an equivalence relation.

(A) Statement–1 is true, Statement–2 is true; Statement–2 is not the correct explanation of Statement–1

(B) Statement–1 is false, Statement–2 is true

(C) Statement–1 is true, Statement–2 is false

(D) Statement–1 is true, Statement–2 is true; Statement–2 is the correct explanation of Statement–1.

MRL117

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	B	C	B	A	B	D	B	D	D

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